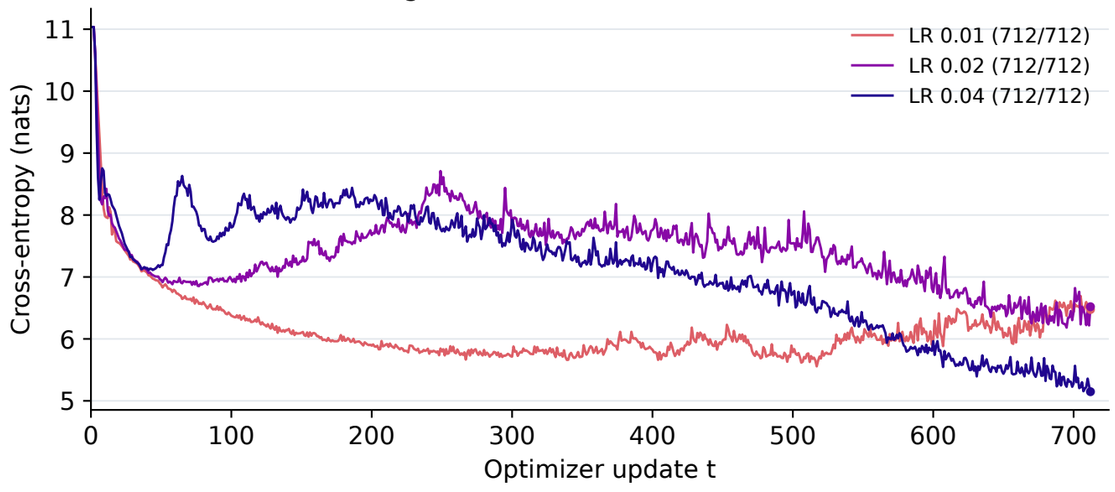
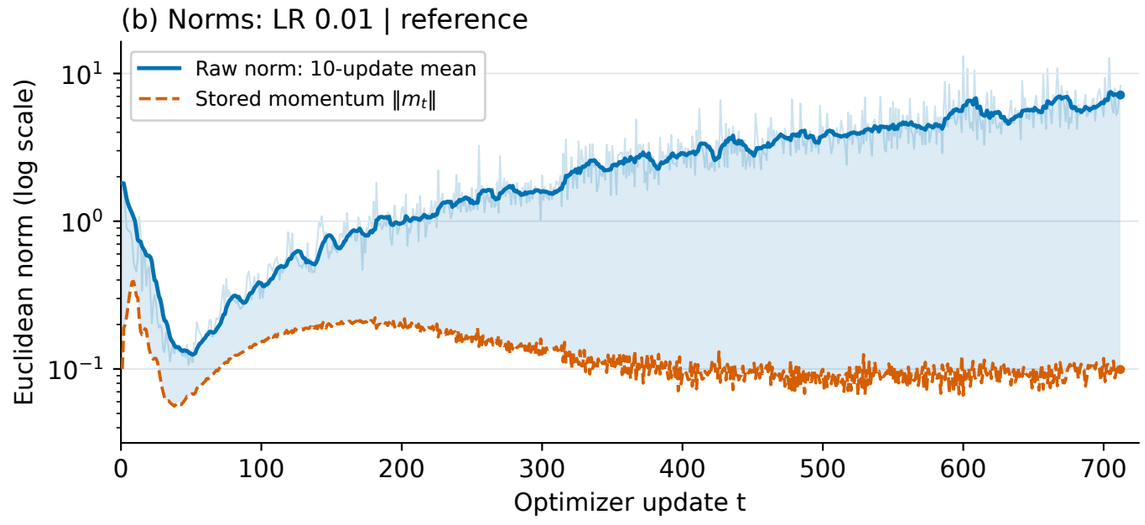
Pythia-14M on DGX Spark: higher learning rates

Matched initialization/order/recipe; new LR 0.02 and 0.04; completed LR 0.01 reference

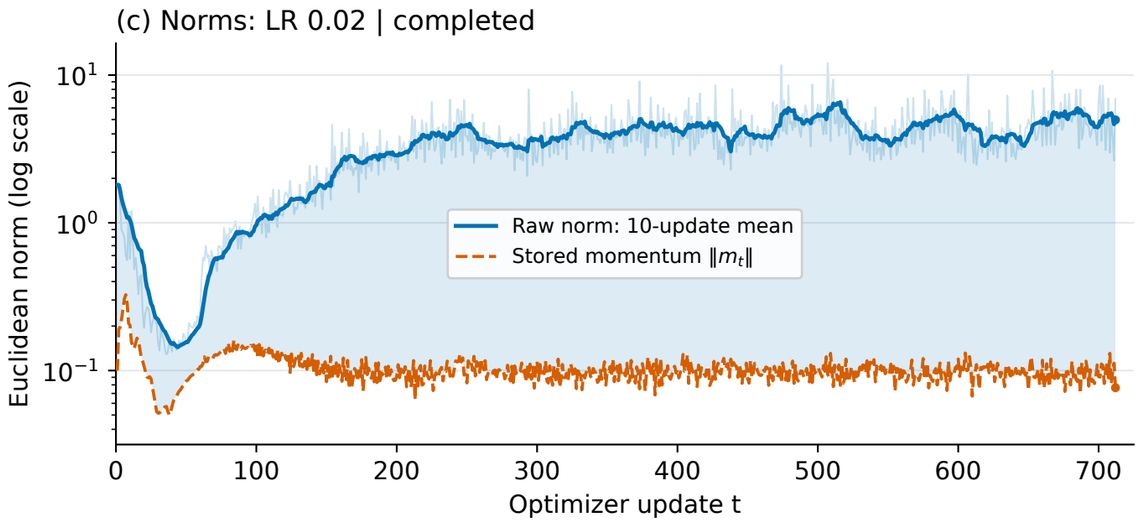
(a) Unsmoothed training loss



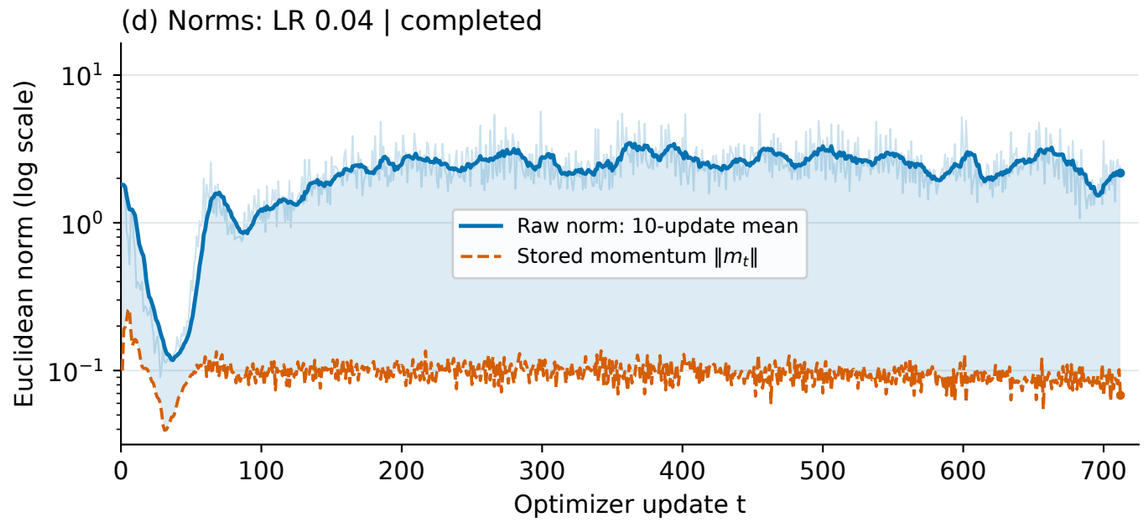
(b) Norms: LR 0.01 | reference



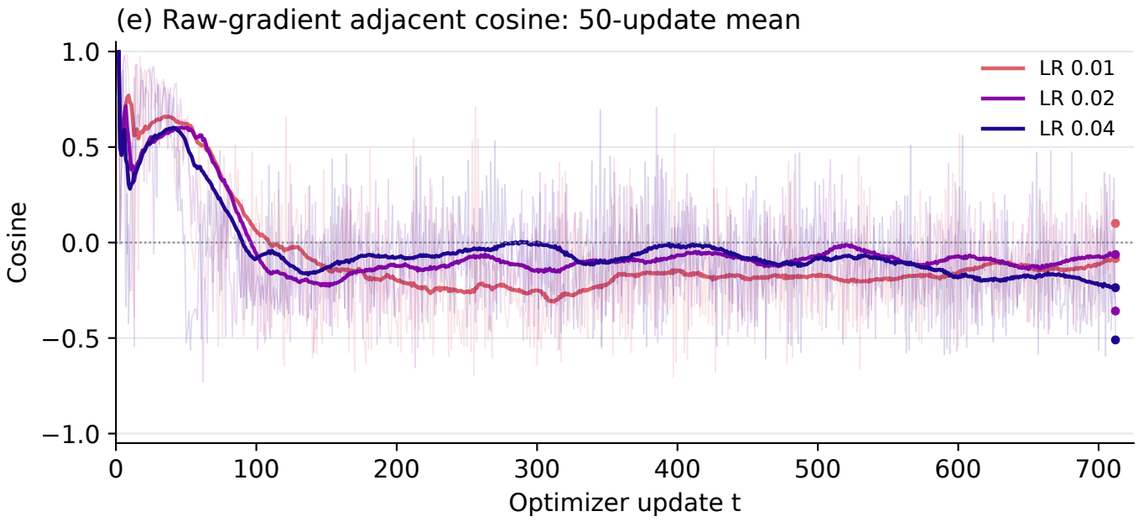
(c) Norms: LR 0.02 | completed



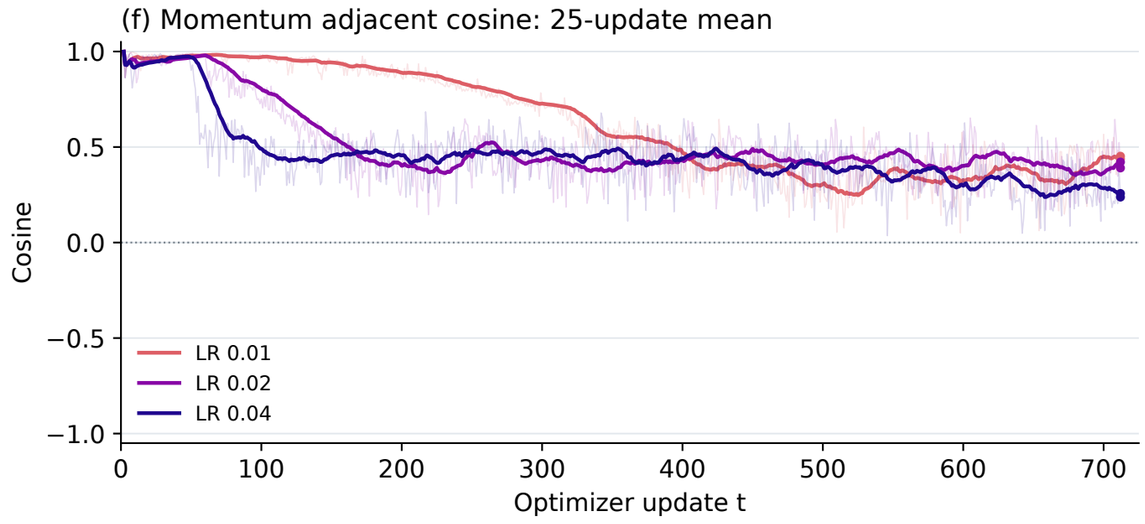
(d) Norms: LR 0.04 | completed



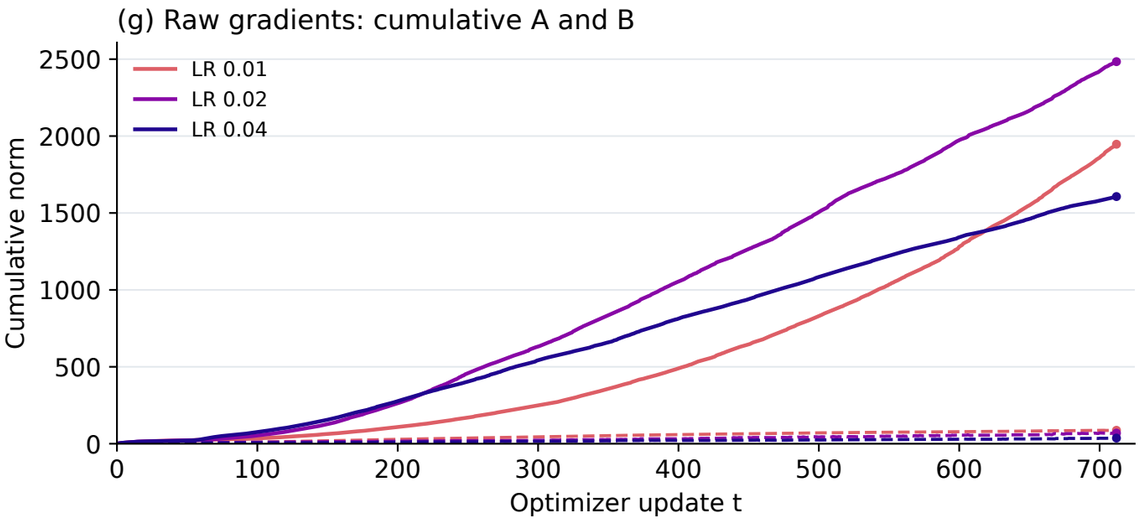
(e) Raw-gradient adjacent cosine: 50-update mean



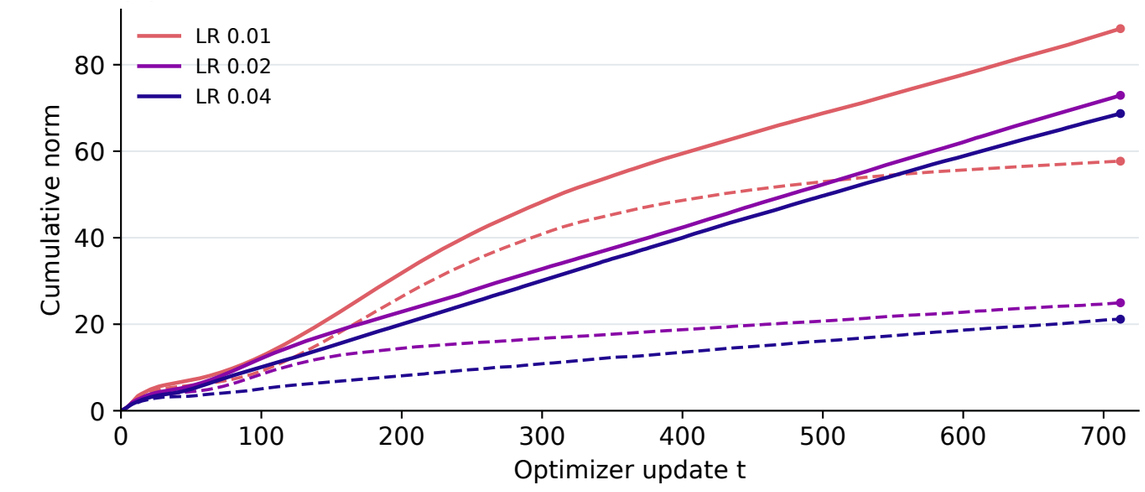
(f) Momentum adjacent cosine: 25-update mean



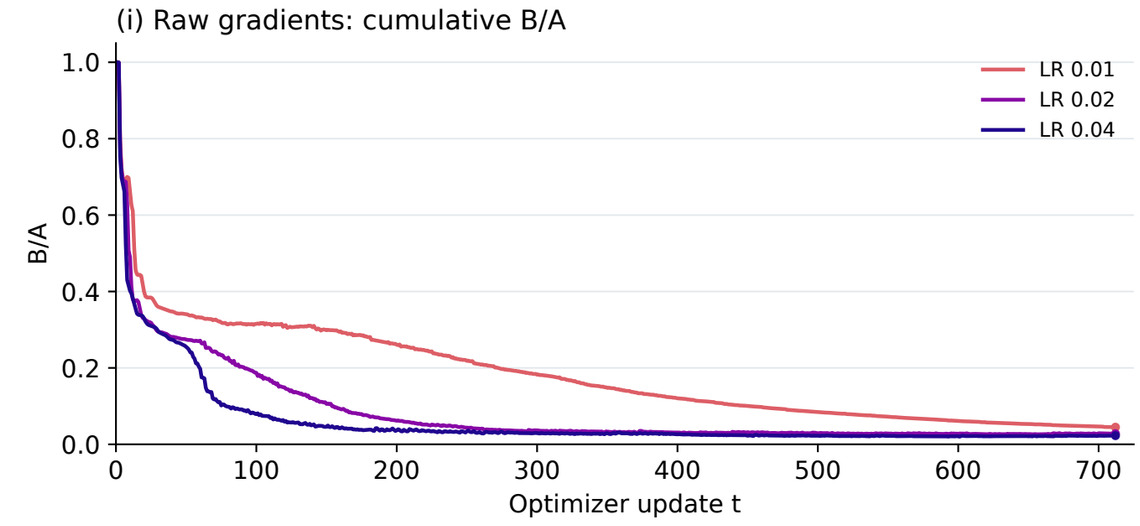
(g) Raw gradients: cumulative A and B



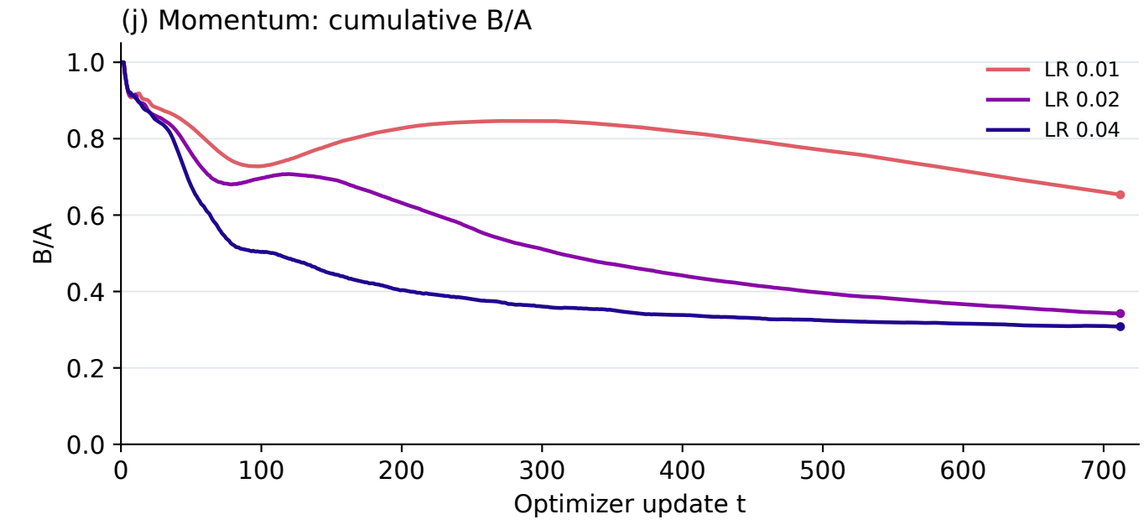
(h) Momentum: cumulative A and B



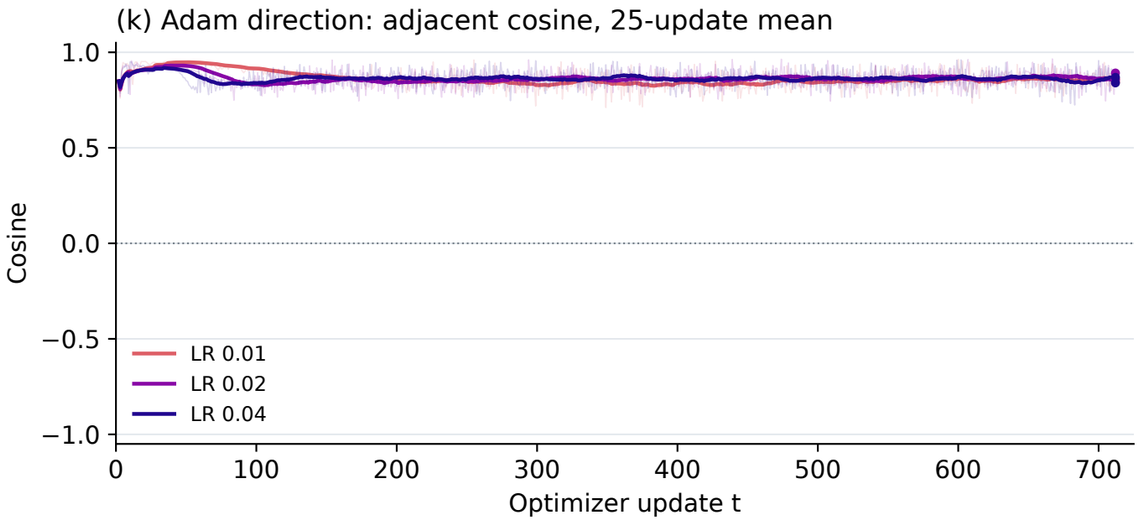
(i) Raw gradients: cumulative B/A



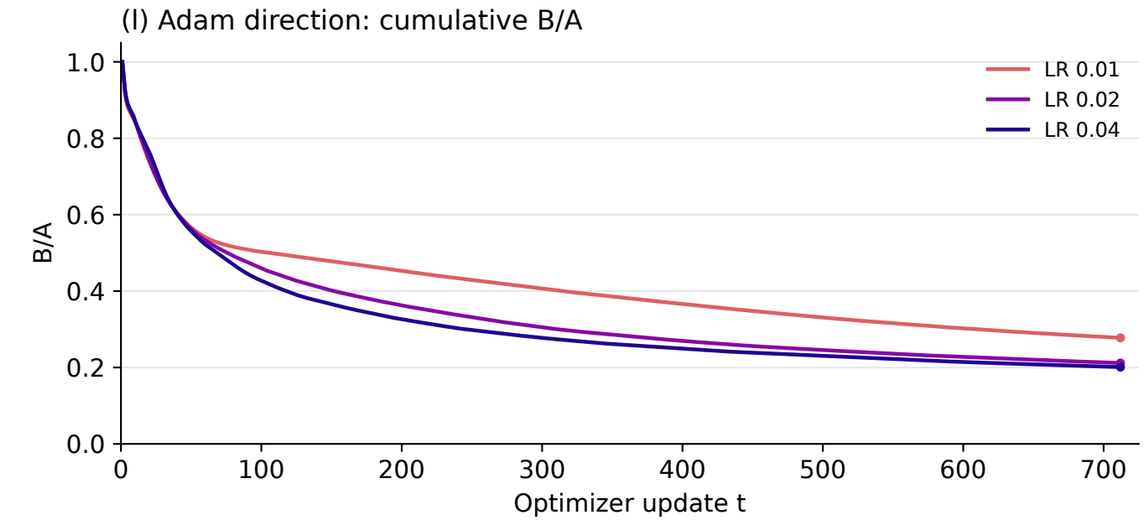
(j) Momentum: cumulative B/A



(k) Adam direction: adjacent cosine, 25-update mean



(l) Adam direction: cumulative B/A



Loss and momentum norms are unsmoothed. Norm panels: faint raw values, trailing 10-update raw mean, fill to momentum (not uncertainty).

Cosine panels: faint originals; raw mean 50 updates, momentum/Adam means 25. Available finite prefixes; undefined first cosine is a gap.

Cumulative geometry is unsmoothed, linear Y: A = sum of vector norms (solid), B = norm of sum (dashed). Circles mark partial update 712.

Raw g is pre-clipping; m is uncorrected Adam exp_avg from clipped g; d excludes LR/decay. One seed; no stability or causal alignment claim.